BASE CHANGE ALONG A CLOSED IMMERSION IS ALMOST COCONTINUOUS FOR THE SYNTOMIC, SMOOTH AND ETALE TOPOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument: it states the precise condition to be verified and obtains it by applying the companion result of the same chapter to the base change of the closed immersion. All of it is kept verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Almost cocontinuous functor between sites. Sites, the definition of an almost cocontinuous functor, which requires that every covering be refined by the images of a covering under the functor. Supplies the condition verified in the proof.
- Preparation of a covering of a closed subscheme. Same chapter, the companion result lifting a syntomic, smooth or etale covering of the closed subscheme to a covering of the ambient scheme with members factoring through it. Supplies the covering with which the almost cocontinuity condition is satisfied.

Source reference. Original definition 04B7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6912–6912. Navigation pointer invoked at proof line 6912; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04E4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6886–6886. Navigation pointer invoked at proof

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line 6925; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a closed immersion of schemes. Let $\tau \in \{syntomic, smooth, étale\}$. The functor $V \mapsto X \times_Y V$ defines an almost cocontinuous functor (see Sites, Definition 04B7) $(Sch/Y)_\tau \to (Sch/X)_\tau$ between big τ sites.*

Proof. We have to show the following: given a morphism $V \to Y$ and any syntomic (resp. smooth, resp. étale) covering $\{U_i \to X \times_Y V\}$, there exists a smooth (resp. smooth, resp. étale) covering $\{V_j \to V\}$ such that for each j, either $X \times_Y V_j$ is empty, or $X \times_Y V_j \to Z \times_Y V$ factors through one of the U_i. This follows on applying Lemma 04E4 above to the closed immersion $X \times_Y V \to V$. □